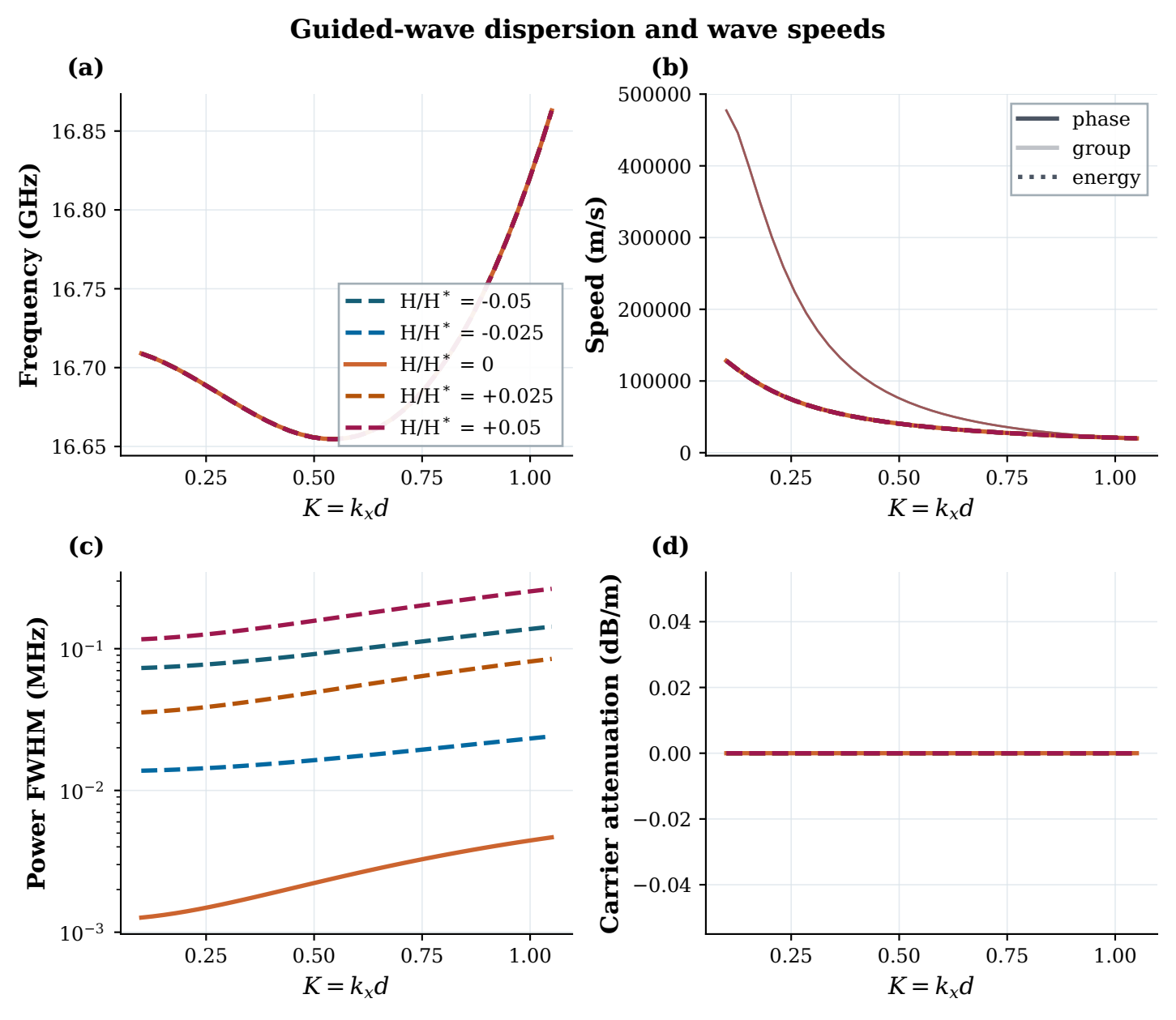
Guided-wave dispersion and wave speeds
(a)
Frequency (GHz)
16.85
16.80
16.75
16.70
16.65
H/H* = -0.05
H/H* = -0.025
H/H* = 0
H/H* = +0.025
H/H* = +0.05
0.25
0.50
0.75
1.00
K = k_x d
(b)
Speed (m/s)
500000
400000
300000
200000
100000
0
phase
group
energy
0.25
0.50
0.75
1.00
K = k_x d
(c)
Power FWHM (MHz)
10^-1
10^-2
10^-3
0.25
0.50
0.75
1.00
K = k_x d
(d)
Carrier attenuation (dB/m)
0.04
0.02
0.00
−0.02
−0.04
0.25
0.50
0.75
1.00
K = k_x d